Chap-3 Continue...

- first rearrange the input element in bit-reversed order
- builds the output transform

Fast Fourier Transform (FFT)

The Fast Fourier Transform is a discrete fourier transform algorithm which reduces the number of computations needed for N points from $2N^2$ to $2N \log_2 N$.

An FFT rapidly computes the transformations by factorizing DFT matrix into a product of sparse (mostly zero) factors.

As a result, it manages to reduce the complexity of computing the DFT from $O(n^2)$ to $O(n \log_2 n)$, when n is the data size.

The FFT algorithm generally fall into two classes:
- decimation in time
- decimation in frequency.

Cooley-Tukey FFT algorithm

The basic idea for FFT (decimation in time) is to break up a transform of length N into two transforms of length N/2 using the identity as:

$$\sum_{n=0}^{N-1} a_n e^{\frac{-2\pi i n k}{N}} = \sum_{n=0}^{N/2-1} a_{2n} e^{\frac{-2\pi i (2n)k}{N}} + \sum_{n=0}^{N/2-1} a_{2n+1} e^{\frac{-2\pi i (2n+1)k}{N}}$$

$$= \sum_{n=0}^{N/2-1} a_n^{even} e^{-\frac{2\pi i k}{N/2}} + \left(\sum_{n=0}^{N/2-1} a_n^{odd} e^{-\frac{2\pi i k}{N/2}}\right) e^{-\frac{2\pi i k}{N}}$$

i.e, $X_k = E_k + e^{-\frac{2\pi}{N}ik} O_k$

It is sometime called Danielson Lanczos Lemma.

The easiest way to visualize this procedure is perhaps via the Fourier matrix.

Applications

(i) Fast large integer & polynomial multiplication

(ii) Filtering algorithms

(iii) Solving difference equations

(iv) Efficient matrix - vector multiplication

FFT (Fast Fourier Transform)
(On the perspective of Image Enhancement)

V.Imp # Redundancy (CH# 5)

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

In digital image processing, three basic ~~data~~ redundancy can be identified and exploited as follows:

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

1) Coding Redundancy We know that average number of bits required to represent each pixel is given by:

$$L_{avg} = \sum_{k=0}^{L-1} l(k)\, p(k)$$

Hence, the number of bits required to represent the image is $n \cdot L_{avg}$ where, n is the total no. of pixel in given image.

Maximum compression

ratio is achieved when Lavg is minimized. But, coding the gray level in such a way that Lavg is not minimized then resulting image is said containing coding redundancy.

2) Interpixel Redundancy: It is related to interpixel correlation within an image. Usually the value of certain pixel in the image can be reasonablly predicted from the values of its neighbours in the image.

Thus, the values of the individual pixel carries relatively small amount of information & much more information about pixel value that can be inferred on the basis of its neighbour's value.

These dependencies between pixel value in the image is called interpixel redundancy.

3) Psychovisual Redundancy: The eye does not respond with equal sensitivity to all visual information.

Human perception

such as for important features (edges, textures) and does not perform quantitative analysis of every pixel in the image.

So, psychovisual redundancy takes into perception of the human visual system.